2026 NBA Finals Analytics Dashboard

An End-to-End Basketball Data Analytics Project Using Python and Power BI

Prepared By

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Project Type

Sports Analytics / Data Visualization

Tools

Python • Pandas • NBA API • Power BI • DAX

Executive Summary

This project analyzes the 2026 NBA Finals between the New York Knicks and the San Antonio Spurs using Python, NBA data, and Power BI. The goal of the project was to build an end-to-end sports analytics workflow that transformed raw basketball data into meaningful team, player, and series-level insights.

The project began with collecting and cleaning NBA game and player statistics before organizing the data into structured analytical tables. The cleaned data was then imported into Power BI, where relationships, calculated columns, and DAX measures were created to support four interactive dashboard pages: Series Overview, Team Analytics, Player Analytics, and Key Insights.

The analysis showed that New York won the series 4-1 behind stronger shooting efficiency and possession control. The Knicks produced their largest winning margins in Game 1 and Game 5, while Jalen Brunson led the offense with 163 total points and 32.6 points per game. The final dashboard provides a clear view of how the series developed, how each team performed, and which players had the greatest impact.

Project Objective

The objective of this project was to develop a complete NBA analytics dashboard to evaluate the outcome of the 2026 NBA Finals from multiple perspectives. Rather than focusing only on final scores, the project was designed to analyze team performance, player production, shooting efficiency, rebounding, turnovers, assists, bench production, and game-by-game momentum.

A second objective was to strengthen my experience working through the full data analytics process. This included collecting data, cleaning and transforming datasets in Python, developing a relational data model, creating DAX measures, designing interactive Power BI dashboards, and communicating the results through clear visualizations and written insights.

Data Collection

The data for this project was collected programmatically using Python and the *nba_api* package. The workflow used NBA game and box score endpoints to retrieve playoff game results and individual player statistics for the 2026 NBA Finals.

The game-level data included each game ID, participating teams, team scores, matchup information, series status, and game winner. Player-level box score data included statistics such as points, rebounds, assists, steals, blocks, turnovers, field goals, three-point shooting, free throws, minutes played, and plus/minus.

The original datasets were saved in a raw data folder before any transformations were performed. This created a clear separation between the original source data and the cleaned datasets used later in the analysis.

The final raw dataset contained five Finals games and 150 player-game records. Keeping the raw data separate also made the workflow easier to reproduce and troubleshoot as the project developed.

Data Cleaning and Preparation

After collecting the raw NBA data, Python and Pandas were used to clean and prepare the datasets for analysis. Duplicate records were removed, game IDs were standardized as text values, and the player dataset was reduced to the statistics that were most relevant to the project.

The cleaned player dataset retained key variables, including player and team identifiers, minutes, points, rebounds, assists, steals, blocks, turnovers, shooting statistics, offensive and defensive rebounds, personal fouls, and plus/minus.

Additional game-level fields were also created to make the series easier to analyze. These included the game number, point differential, winning and losing scores, close-game classification, cumulative series wins, and the series leader after each game.

Feature engineering was then used to create more advanced basketball metrics.

These included:

- Effective Field Goal Percentage
- True Shooting Percentage
- Assist-to-Turnover Ratio
- Three-Point Attempt Rate
- Free Throw Rate
- Points Per Minute
- Rebounds Per Minute
- Assists Per Minute
- Stocks, calculated as steals plus blocks

- Player Efficiency

These calculations expanded the analysis beyond traditional box-score statistics and enabled evaluation of player performance through both production and efficiency.

Team-level datasets were also created by aggregating player statistics for each team and game. From these totals, additional metrics such as Team Effective Field Goal Percentage, Team True Shooting Percentage, Team Assist-to-Turnover Ratio, Team Free Throw Rate, and Team Efficiency were calculated.

By the end of the preparation stage, separate player, team, and game analytics datasets had been created for use in the final Power BI dashboard.

Data Modeling

To support the Power BI analysis, the cleaned data was organized into a relational model rather than relying on one large flat table.

The model included three dimension tables and a central player statistics fact table:

dim_player contained unique player information, including player ID, player name, team ID, and team abbreviation.

dim_team contained the two Finals teams and their unique team IDs.

dim_game contained the five games in the series along with matchup, series state, and winner information.

fact_player_stats contained the numerical player performance records for each game, including points, rebounds, assists, shooting statistics, turnovers, defensive statistics, minutes, and plus/minus.

The fact table contained 150 player-game records and was connected to the dimension tables through identifiers such as Player ID, Team ID, and Game ID.

In addition to the core model, separate player, team, and game analytics tables were used to support dashboard calculations and visualizations. This structure made it easier to filter the report by player, team, and game while keeping the analytics logic organized.

Organizing the project this way also helped reinforce the difference between descriptive information, such as player and team names, and measurable performance data used for calculations.

Power BI Dashboard Development

After the data was cleaned, transformed, and organized, the final datasets were imported into Power BI to build an interactive analytics dashboard for the 2026 NBA Finals.

The goal of the dashboard was to make the series easy to understand at multiple levels. Instead of placing every metric on a single page, the report was divided into four dashboard pages: Series Overview, Team Analytics, Player Analytics, and Key Insights. Each page was designed to answer a different type of question about the Finals.

The Series Overview provides a high-level summary of the matchup and series outcome. It gives users a quick understanding of how the series progressed, which team won, and how individual games contributed to the final result.

The Player Analytics page allows users to evaluate individual performances throughout the series. Interactive filters were added for player, team, game, and player role, allowing the dashboard to move from an overall series view to more specific player-level analysis. Visuals were also designed to highlight leading scorers, efficiency, minutes played, and other areas of player production.

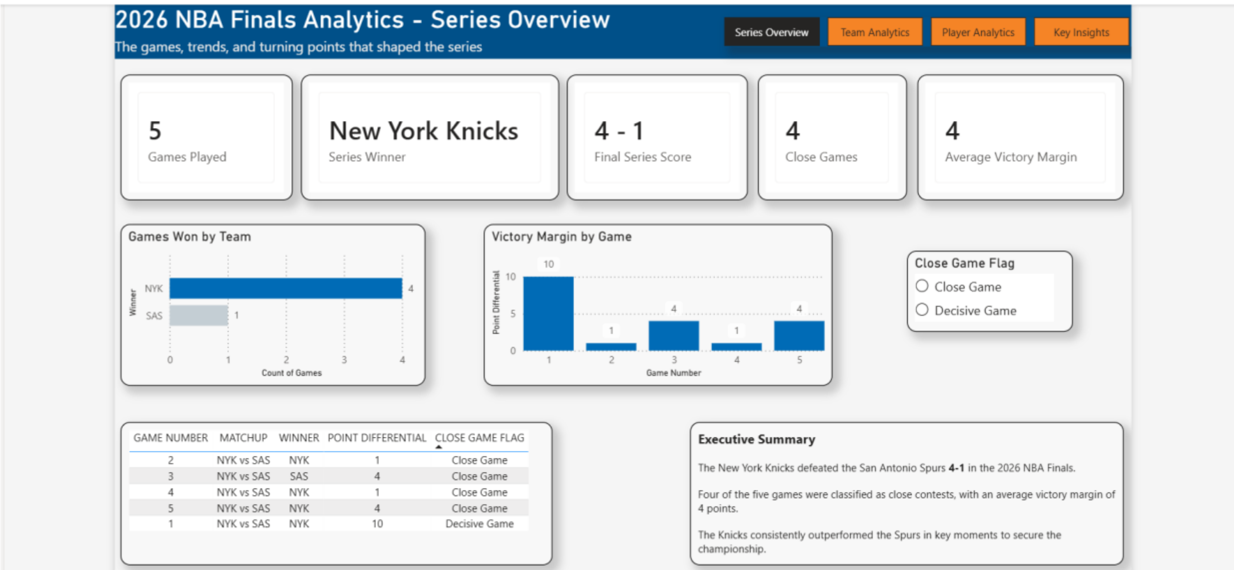
The Key Insights page brings the analysis together by presenting the most important findings from the series. Rather than requiring users to interpret every individual chart, this page highlights major performance trends, standout players, game margins, and the factors that contributed most to the final outcome.

DAX measures were created throughout the dashboard to calculate metrics dynamically based on user selections. Measures included calculations for scoring, shooting efficiency, rebounding, turnovers, player performance, game margins, and other series-level statistics. Using measures instead of relying only on pre-calculated columns allowed the dashboard to respond to filters and slicers more effectively.

Interactive slicers and visual interactions were also used so slicers could filter the dashboard by team, player, or game. This made it possible to explore the Finals from different perspectives without creating separate reports for every comparison.

The final dashboard was designed to balance analytical depth with readability. Key performance indicators were displayed prominently through cards, while charts, tables, and supporting visuals were used to provide additional context. Consistent formatting, titles, subtitles, and page organization were applied across the dashboard to create a more professional and cohesive presentation.

Overall, the Power BI development stage transformed the prepared datasets into an interactive analytical product that could communicate both the overall story of the series and the individual statistics behind it.



Series Overview Analysis

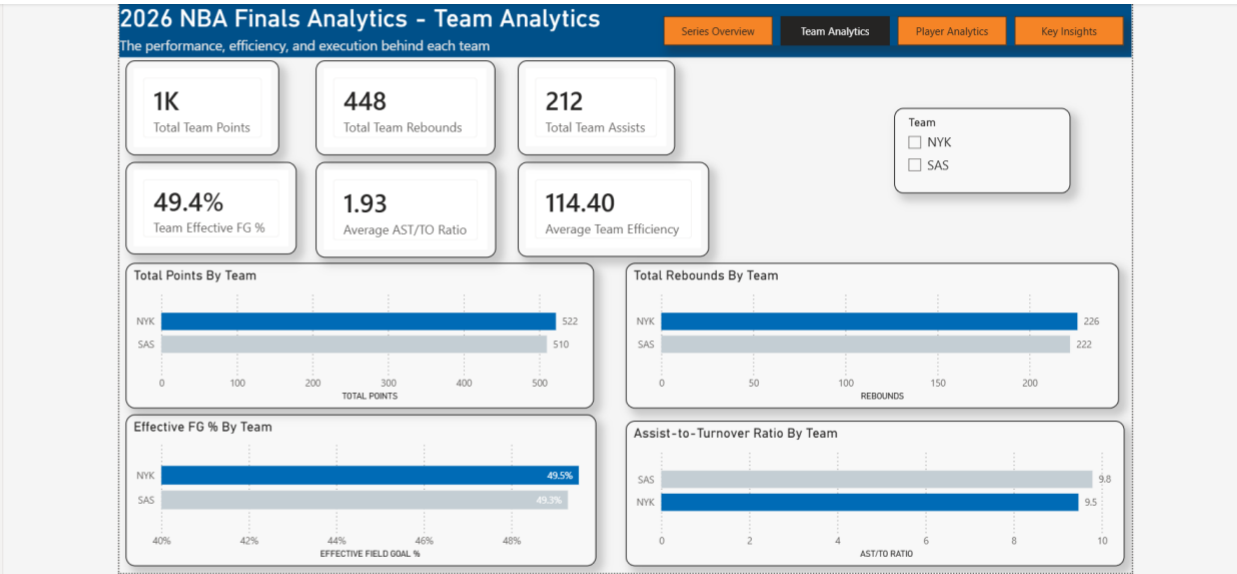
The Series Overview dashboard provides a high-level summary of how the 2026 NBA Finals developed between the New York Knicks and the San Antonio Spurs. The purpose of this page was to give the user an immediate understanding of the series results before moving into more detailed team and player analysis.

New York won the series 4-1, establishing control over the matchup across the five games. The dashboard highlights the outcome of each game, the series progression, and the point differential between the two teams. This makes it easier to identify when New York created separation and how San Antonio responded throughout the finals.

One of the clearest trends was New York's ability to create larger scoring margins in key games. The Knicks recorded their largest margin of victory in Games 1 and 5, allowing them to begin the series strongly and eventually close it out with another decisive performance. Looking at game margins alongside the series progression helped show that the championship result was not based on a single isolated performance but on New York maintaining a stronger overall level of play throughout the series.

The page also serves as the starting point for the rest of the dashboard. While the Series Overview explains what happened, the following Team Analytics and Player Analytics pages provide more detail on why it happened by examining shooting efficiency, possession control, scoring production, rebounding, turnovers, and individual player performances.

Overall, the Series Overview establishes the main story of the Finals: New York controlled the series and ultimately won 4-1, while the remaining dashboard pages break down the statistical factors and individual performances that contributed to that result.



Team Analytics

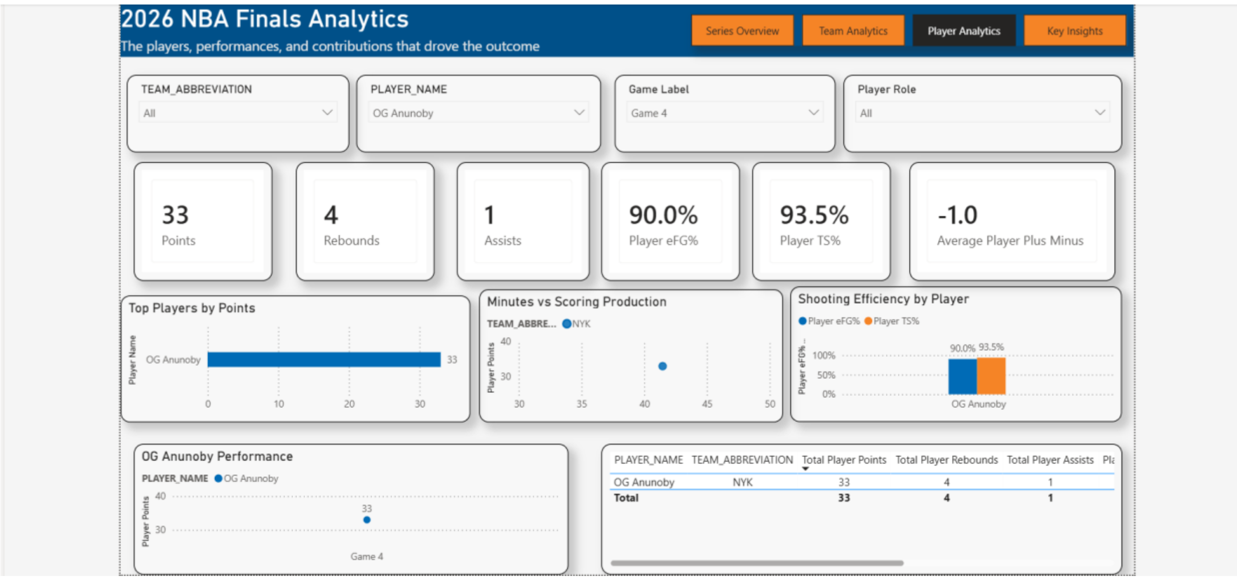
The Team Analytics dashboard provides a more detailed comparison of the New York Knicks and San Antonio Spurs by focusing on the team-level statistics that influenced the series. This page was designed to move beyond the final scores and help explain the differences in how each team performed throughout the Finals.

The dashboard compares the two teams across key areas, including points, rebounds, assists, shooting efficiency, team efficiency, and assist-to-turnover ratio. These metrics provide a clearer view of how effectively each team scored, created opportunities for teammates, controlled possessions, and used their offensive possessions.

One of the main takeaways from the analysis was New York's stronger overall efficiency. The Knicks combined strong scoring with better shooting and possession control, which helped them maintain an advantage throughout the series. San Antonio remained competitive in several statistical areas, but New York was more consistent in converting its opportunities into points.

The team comparison also shows why looking only at total scoring does not tell the full story of the match. Metrics such as Effective Field Goal Percentage and Assist-to-Turnover Ratio provide additional context by showing how efficiently each team created scoring opportunities and protected the basketball.

Overall, the Team Analytics dashboard helps explain why New York won the series 4-1. Their advantage came from a combination of scoring, efficiency, and execution rather than one single statistic. This provides the foundation for the Player Analytics section, which examines the individual performances that contributed to those team-level results.



Player Analytics

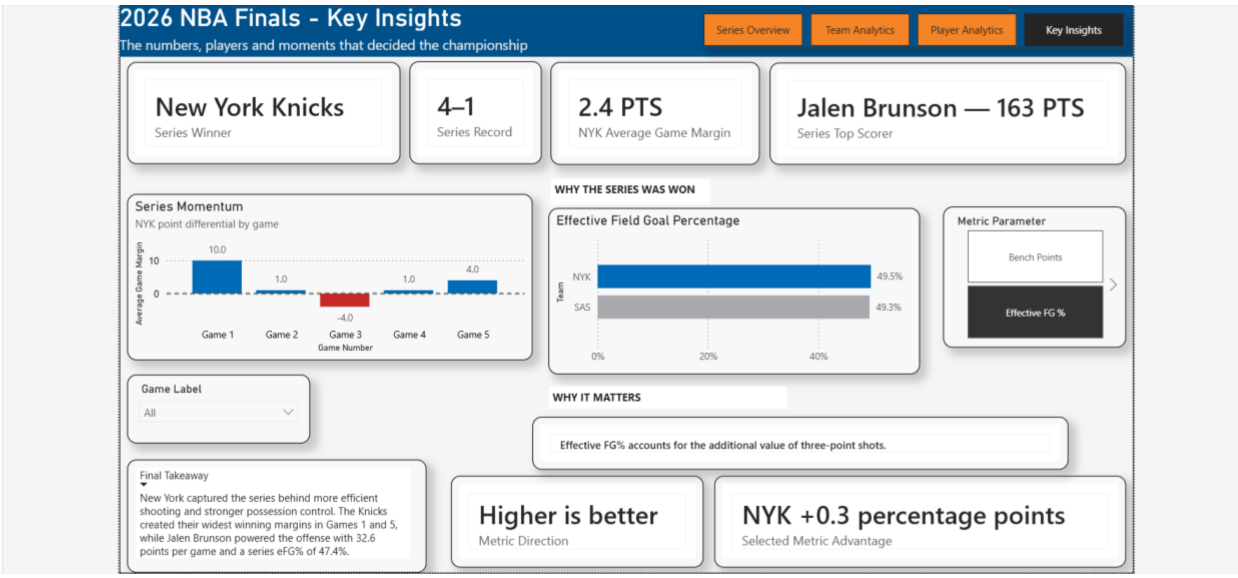
The Player Analytics dashboard focuses on the individual performances that contributed to the series' overall results. While the previous section compared New York and San Antonio at the team level, this page provides a closer look at how individual players performed across the five games.

The dashboard allows users to compare players using statistics such as points, rebounds, assists, minutes played, shooting efficiency, and overall player efficiency. Interactive filters also let users narrow the analysis by player, team, game, or player role, allowing them to explore specific performances throughout the series.

One of the biggest standouts was Jalen Brunson, who led the series in total points and averaged 32.6 points per game, anchoring New York's offense throughout the Finals. Alongside him, OG Anunoby delivered a crucial Game 4 performance, providing elite two-way impact on both ends of the floor and helping swing momentum in a pivotal matchup that ultimately played a key role in the series outcome.

The dashboard also shows that player impact goes beyond scoring. Rebounds, assists, shooting efficiency, minutes played, and other contributions provide additional context when comparing players. This makes it possible to identify players who contributed in different ways even if they were not among the leading scorers.

Overall, the Player Analytics dashboard helps connect individual performance to the larger team results. By examining both production and efficiency, the analysis provides a clearer understanding of which players had the greatest impact and how their performance contributed to the outcome of the series.



Key Insights

The Key Insights dashboard brings together the most important findings from the analysis and provides a final explanation of why the New York Knicks were able to win the 2026 NBA Finals. This page combines the series outcome, game-by-game momentum, team efficiency, and individual performance into one summary view.

New York won the series 4-1 with an average game margin of 2.4 points. The Series Momentum chart shows that the Knicks created their largest advantage in Game 1, with a 10-point margin, and finished the series with a 4-point win in Game 5. Although San Antonio won Game 3, New York responded and maintained control of the series.

The “Why the Series Was Won” section highlights Effective Field Goal Percentage, where New York held a slight advantage of 49.5% compared with San Antonio’s 49.3%. While the difference was small, it shows that the Knicks were slightly more efficient when accounting for the additional value of three-point shots. Combined with stronger possession control, this helped New York create an advantage over the course of the series.

Individual performance was also a major factor. Jalen Brunson led all scorers with 163 total points and averaged 32.6 points per game, making him the main offensive contributor for New York throughout the Finals.

Overall, the dashboard shows that New York's championship was not the result of one single statistic. The Knicks combined efficient shooting, possession control, strong individual performances, and the ability to win important games, ultimately allowing them to close out the series 4-1.

Challenges and Solutions

One of the main challenges in this project was transforming raw NBA data into a structure that could be used effectively in Power BI. The original game and player datasets contained many fields, so the data had to be cleaned and reduced to the statistics that were most useful for the analysis. Python and Pandas were used to remove unnecessary columns, standardize identifiers, and create additional variables such as game number, point differential, and series status.

Another challenge was developing metrics that went beyond basic box-score totals. Measures such as Effective Field Goal Percentage, True Shooting Percentage, Assist-to-Turnover Ratio, and player efficiency required additional calculations. These metrics were created during the data preparation stage and through DAX so the dashboard could provide a deeper view of team and player performance.

The data model was also important because the project included player, team, and game-level information. Organizing the data into fact and dimension tables helped reduce duplication and made filtering by player, team, and game more reliable.

A final challenge was dashboard design. Early versions of the report had issues with spacing, visual consistency, and the ability to fit multiple metrics on one page. These problems were improved by reorganizing visuals into cleaner layouts, using consistent titles and formatting, and separating the analysis into four focused dashboard pages.

Overall, each challenge helped improve the final project and strengthened the workflow from data collection and cleaning to modeling, visualization, and analysis.

Limitations

Although the project provides a detailed analysis of the 2026 NBA Finals, there are several limitations to consider. The biggest limitation is the small sample size, as the series had only five games. This means that one strong or weak performance can have a large effect on averages and overall results.

The analysis also relies mainly on traditional box score and efficiency statistics. These metrics are useful, but they don't fully capture areas such as defensive positioning, shot difficulty, player matchups, off-ball movement, or coaching adjustments.

The dashboard also focuses only on the Finals and doesn't include regular-season or earlier playoff performance. Adding that information could provide more context about whether certain trends were unique to the Finals or consistent throughout the regular season.

Overall, the project provides a strong summary of the series, but future versions could include more games, track more data points, and add advanced metrics to create a deeper analysis.

Conclusion

This project demonstrated the full process of turning raw NBA data into a complete analytics dashboard and written analysis. Using Python, Pandas, the NBA API, Power BI, and DAX, the project moved from data collection and cleaning to feature engineering, data modeling, visualization, and interpretation.

The final analysis showed that the New York Knicks won the 2026 NBA Finals 4-1 by combining strong individual performances with efficient team play and consistent execution. Jalen Brunson led the series in scoring, while New York also held advantages in shooting efficiency and possession control.

Beyond basketball results, this project helped strengthen my ability to work through an end-to-end analytics workflow. It required both technical skills and decision-making, especially when choosing useful metrics, organizing the data model, and designing easy-to-understand dashboards.

Overall, the project shows how sports data can be transformed into clear, meaningful insights that explain not only what happened in a series but also the factors that contributed to the final outcome.

Tools and Technologies

This project used several tools throughout the data analytics process, with each one serving a different purpose.

Python was used for data collection, cleaning, transformation, and feature engineering. It helped automate the workflow and prepare the raw NBA data for analysis.

Pandas was used to organize datasets, remove unnecessary fields, create new variables, calculate advanced metrics, and build the final analytical tables.

NBA API was used as the main data source for retrieving game results and player box score statistics.

Power BI was used to build the final interactive dashboard, organize the data model, create visualizations, and present the results across the Series Overview, Team Analytics, Player Analytics, and Key Insights pages.

DAX was used inside Power BI to create dynamic measures and calculations that respond to filters and slicers.

GitHub was used to organize and document the project files, code, datasets, and final portfolio materials.

Together, these tools supported the complete workflow from raw data collection to the final dashboard and written analysis.